

```
CREATE DATABASE University;  
use University;
```

```
CREATE TABLE Student (  
    StudentID INT PRIMARY KEY,  
    FirstName VARCHAR(50),  
    LastName VARCHAR(50),  
    Email VARCHAR(100) UNIQUE  
);
```

```
CREATE TABLE Instructor (  
    InstructorID INT PRIMARY KEY,  
    InstructorName VARCHAR(100),  
    Email VARCHAR(100) UNIQUE  
);
```

```
CREATE TABLE Course (  
    CourseID INT PRIMARY KEY,  
    CourseName VARCHAR(100),  
    InstructorID INT,  
    FOREIGN KEY (InstructorID) REFERENCES Instructor(InstructorID)  
);
```

```
CREATE TABLE Enrollment (  
    EnrollmentID INT AUTO_INCREMENT PRIMARY KEY,  
    StudentID INT,  
    CourseID INT,  
    EnrollmentDate DATE,  
    FOREIGN KEY (StudentID) REFERENCES Student(StudentID),  
    FOREIGN KEY (CourseID) REFERENCES Course(CourseID)  
);
```

```
CREATE TABLE Assignment (  
    AssignmentID INT PRIMARY KEY,  
    AssignmentName VARCHAR(100),  
    CourseID INT,  
    DueDate DATE,  
    FOREIGN KEY (CourseID) REFERENCES Course(CourseID)  
);
```

```
129      -- Show all students
130  •    SELECT * FROM Student;
131
```

Result Grid



Filter Rows:

Edit:



	StudentID	FirstName	LastName	Email
▶	1	First1	Last1	student1@school.edu
	2	First2	Last2	student2@school.edu
	3	First3	Last3	student3@school.edu
	4	First4	Last4	student4@school.edu
	5	First5	Last5	student5@school.edu
	6	First6	Last6	student6@school.edu
	7	First7	Last7	student7@school.edu
	8	First8	Last8	student8@school.edu
	9	First9	Last9	student9@school.edu
	10	First10	Last10	student10@school.edu
	11	First11	Last11	student11@school.edu
	12	First12	Last12	student12@school.edu
	13	First13	Last13	student13@school.edu
	14	First14	Last14	student14@school.edu
	15	First15	Last15	student15@school.edu
	16	First16	Last16	student16@school.edu
	17	First17	Last17	student17@school.edu
	18	First18	Last18	student18@school.edu
	19	First19	Last19	student19@school.edu
	20	First20	Last20	student20@school.edu
●	NULL	NULL	NULL	NULL

```
132      -- Show all instructors
133  •    SELECT * FROM Instructor;
```

Result Grid	Filter Rows:	Edit:
InstructorID	InstructorName	Email
1	Instructor 1	instructor1@school.edu
2	Instructor2	instructor2@school.edu
3	Instructor3	instructor3@school.edu
4	Instructor4	instructor4@school.edu
5	Instructor5	instructor5@school.edu
NULL	NULL	NULL

```
135      -- Show all courses
136 •     SELECT * FROM Course;
```

Result Grid |   Filter Rows:

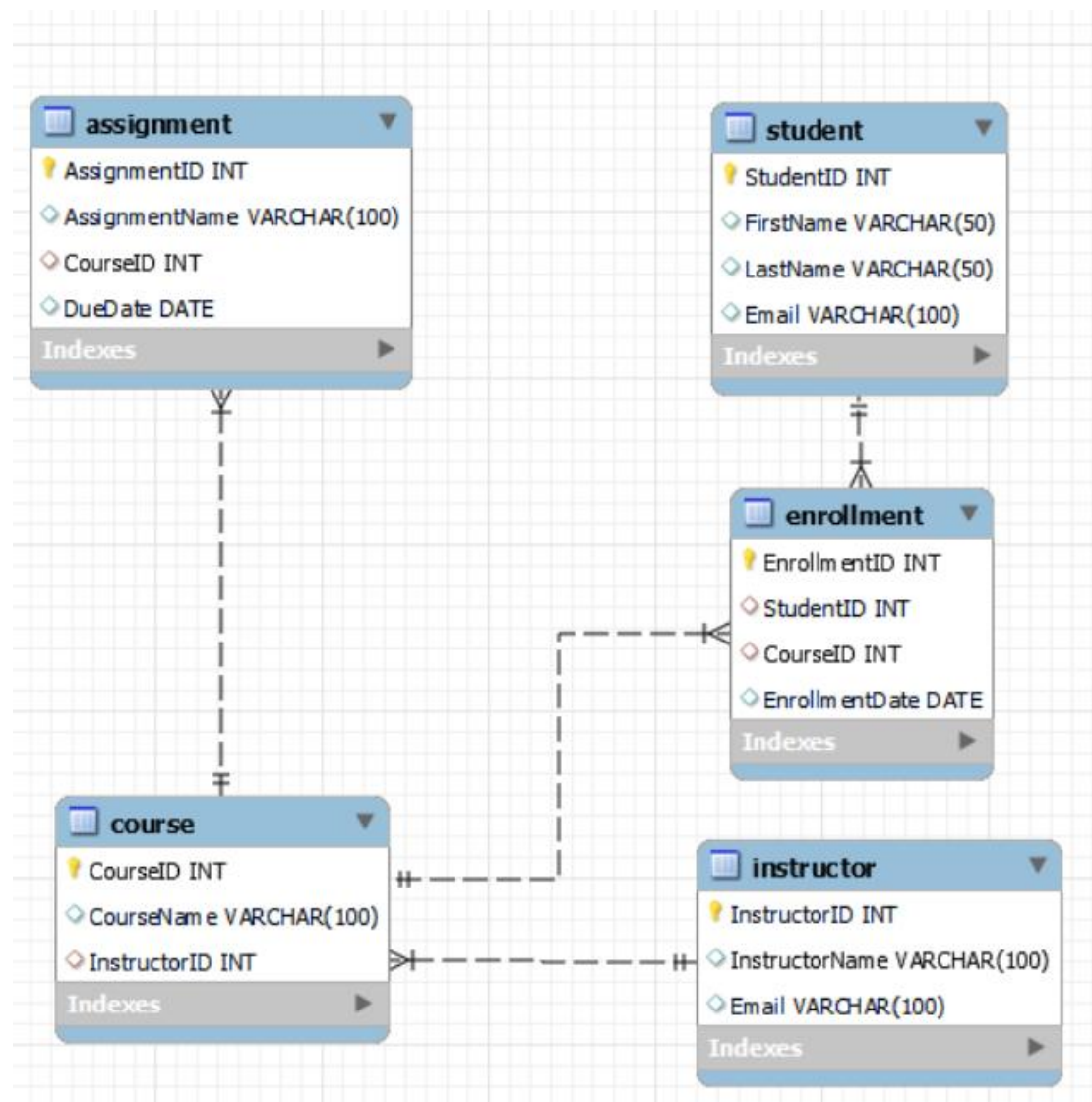
	CourseID	CourseName	InstructorID
▶	1	Course 1	3
	2	Course2	3
	3	Course3	3
	4	Course4	5
	5	Course5	2
	6	Course6	5
	7	Course7	5
	8	Course8	5
	9	Course9	3
	10	Course 10	3
•	NULL	NULL	NULL

```
138 -- Show all enrollments
139 • SELECT * FROM Enrollment;
```

	EnrollmentID	StudentID	CourseID	EnrollmentDate
▶	1	10	6	2023-02-17
	2	19	3	2023-04-23
	3	6	8	2023-06-27
	4	19	3	2023-08-26
	5	12	4	2023-09-22
	6	5	7	2023-05-15
	7	15	5	2023-07-23
	8	11	5	2023-08-28
	9	19	1	2023-10-20
	10	5	6	2023-11-01
	11	3	10	2023-03-30
	12	15	6	2023-11-30
	13	20	10	2023-04-13
	14	10	5	2023-05-05
	15	13	6	2023-05-31
	16	11	7	2023-12-05
	17	4	4	2023-01-21
	18	6	9	2023-03-10
	19	17	7	2023-11-14
	20	4	5	2023-05-14
▲	NULL	NULL	NULL	NULL

```
141      -- Show all assignments
142 •    SELECT * FROM Assignment;
```

	AssignmentID	AssignmentName	CourseID	DueDate
▶	1	Assignment1	5	2024-04-07
	2	Assignment2	10	2024-04-09
	3	Assignment3	4	2024-02-16
	4	Assignment4	4	2024-05-26
	5	Assignment5	8	2024-03-25
	6	Assignment6	8	2024-03-07
	7	Assignment7	5	2024-01-05
	8	Assignment8	4	2024-03-16
	9	Assignment9	6	2024-05-31
	10	Assignment10	1	2024-03-01
	11	Assignment11	4	2024-02-16
	12	Assignment12	6	2024-06-12
	13	Assignment13	3	2024-02-14
	14	Assignment14	4	2024-06-30
	15	Assignment15	5	2024-06-12
	16	Assignment16	2	2024-01-30
	17	Assignment17	6	2024-06-26
	18	Assignment18	10	2024-01-19
	19	Assignment19	3	2024-01-14
	20	Assignment20	10	2024-03-15
•	NULL	NULL	NULL	NULL



schoolmsdb_ER university_sample_data bankDB

Limit to 5000 rows

```

154 SELECT
155     s.StudentID,
156     CONCAT(s.FirstName, ' ', s.LastName) AS StudentName,
157     c.CourseName,
158     i.InstructorName,
159     e.EnrollmentDate
160 FROM Enrollment e
161 JOIN Student s ON e.StudentID = s.StudentID
162 JOIN Course c ON e.CourseID = c.CourseID
163 JOIN Instructor i ON c.InstructorID = i.InstructorID;
  
```

StudentID	StudentName	CourseName	InstructorName	EnrollmentDate
19	First19 Last19	Course1	Instructor3	2023-10-20
19	First19 Last19	Course3	Instructor3	2023-04-23
19	First19 Last19	Course3	Instructor3	2023-08-26
12	First12 Last12	Course4	Instructor5	2023-09-22
4	First4 Last4	Course4	Instructor5	2023-01-21
15	First15 Last15	Course5	Instructor2	2023-07-23
11	First11 Last11	Course5	Instructor2	2023-08-28
10	First10 Last10	Course5	Instructor2	2023-05-05
4	First4 Last4	Course5	Instructor2	2023-05-14
10	First10 Last10	Course6	Instructor5	2023-02-17
5	First5 Last5	Course6	Instructor5	2023-11-01
15	First15 Last15	Course6	Instructor5	2023-11-30
13	First13 Last13	Course6	Instructor5	2023-05-31
5	First5 Last5	Course7	Instructor5	2023-05-15
11	First11 Last11	Course7	Instructor5	2023-12-05

165 -- =====

166 -- STEP 10: ASSIGNMENT + COURSE VIEW

167 -- =====

168

169 • SELECT

170 a.AssignmentName,

171 c.CourseName,

172 a.DueDate

173 FROM Assignment a

174 JOIN Course c ON a.CourseID = c.CourseID;

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

AssignmentName	CourseName	DueDate
Assignment10	Course1	2024-03-01
Assignment16	Course2	2024-01-30
Assignment13	Course3	2024-02-14
Assignment19	Course3	2024-01-14
Assignment3	Course4	2024-02-16
Assignment4	Course4	2024-05-26
Assignment8	Course4	2024-03-16
Assignment11	Course4	2024-02-16
Assignment14	Course4	2024-06-30
Assignment1	Course5	2024-04-07
Assignment7	Course5	2024-01-05
Assignment15	Course5	2024-06-12
Assignment9	Course6	2024-05-31
Assignment12	Course6	2024-06-12
Assignment17	Course6	2024-06-26

182 -- =====

183 -- A: ATOMICITY TEST

184 -- =====

185 • START TRANSACTION;

186 • INSERT INTO Student VALUES (21, 'Atomic', 'Test', 'atomic@school.edu');

187 • INSERT INTO Enrollment (StudentID, CourseID, EnrollmentDate)

188 VALUES (21, 999, '2024-01-01');

189 -- Invalid CourseID

190 • ROLLBACK;

191 • SELECT * FROM Student WHERE StudentID = 21;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

StudentID	FirstName	LastName	Email
1	IsolationTest	Last1	john.updated@school.edu

194 -- =====

195 -- C: CONSISTENCY TEST

196 -- =====

197

198 • INSERT INTO Student VALUES (22, 'Consistency', 'Test', 'student1@school.edu');

199 -- Expected: FAIL (duplicate email)

200

201 • INSERT INTO Course VALUES (11, 'InvalidCourse', 999);

202 -- Expected: FAIL (invalid instructor)

15 21:08:20 INSERT INTO Course VALUES (11, 'InvalidCourse', 999) Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails ('university`.`course`, CON... 0.016 sec

16 21:08:43 INSERT INTO Student VALUES (22, 'Consistency', 'Test', 'student1@school.edu') Error Code: 1062. Duplicate entry '22' for key 'student.PRIMARY' 0.016 sec

204 -- =====

205 -- I: ISOLATION TEST

206 -- =====

207

208 • START TRANSACTION;

209 • UPDATE Student

210 SET FirstName = 'IsolationTest'

211 WHERE StudentID = 1;

212 -- In another session:

213 • SELECT * FROM Student WHERE StudentID = 1;

214 • ROLLBACK;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

StudentID	FirstName	LastName	Email
1	IsolationTest	Last1	john.updated@school.edu

217 -- D: DURABILITY TEST
218 -- =====
219
220 • START TRANSACTION;
221
222 • INSERT INTO Student VALUES (23, 'Durable', 'Student', 'durable@school.edu');
223
224 • COMMIT;
225
226 • SELECT * FROM Student WHERE StudentID = 23;
227 -- Expected: Permanent

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Contents: |

StudentID	FirstName	LastName	Email
23	Durable	Student	durable@school.edu

229 -- =====
230 -- STEP 11: PRIMARY KEY TEST
231 -- =====
232
233 • INSERT INTO Student VALUES (1, 'DuplicatePK', 'Fail', 'duplicatepk@school.edu');
234 -- Expected: FAIL

29 21:11:53 INSERT INTO Student VALUES (1, 'DuplicatePK', 'Fail', 'duplicatepk@school.edu') Error Code: 1062. Duplicate entry '1' for key 'student.PRIMARY' 0.000 sec

236 -- =====
237 -- STEP 12: FOREIGN KEY TEST
238 -- =====
239
240 • INSERT INTO Enrollment (StudentID, CourseID, EnrollmentDate)
241 VALUES (999, 1, '2024-01-01');
242 -- Expected: FAIL

30 21:12:31 INSERT INTO Enrollment (StudentID, CourseID, EnrollmentDate) VALUES (999, 1, '2024-01-01') Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails ('university'.enrollment', C... 0.000 sec

-- =====
-- STEP 13: UNIQUE TEST
-- =====
• INSERT INTO Instructor VALUES (6, 'DuplicateEmail', 'instructor1@school.edu');
-- Expected: FAIL

31 21:13:15 INSERT INTO Instructor VALUES (6, 'DuplicateEmail', 'instructor1@school.edu') Error Code: 1062. Duplicate entry 'instructor1@school.edu' for key 'Instructor.Email' 0.000 sec

251 -- =====
252 -- STEP 14: AGGREGATE TESTS
253 -- =====
254
255 • SELECT COUNT(*) AS TotalStudents FROM Student;
256 • SELECT COUNT(*) AS TotalCourses FROM Course;
257 • SELECT COUNT(*) AS TotalEnrollments FROM Enrollment;
258 • SELECT COUNT(*) AS TotalAssignments FROM Assignment;

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

TotalStudents
22

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

TotalCourses
10

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

TotalEnrollments
20

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

TotalAssignments
20

260 -- =====

261 -- STEP 15: GROUP BY TEST

262 -- =====

263

264 • `SELECT CourseID, COUNT(StudentID) AS StudentCount`

265 `FROM Enrollment`

266 `GROUP BY CourseID;`

267

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

CourseID	StudentCount
1	1
3	2
4	2
5	4
6	4
7	3
8	1
9	1
10	2

268 -- =====

269 -- STEP 16: ORDER BY TEST

270 -- =====

271

272 • `SELECT * FROM Student`

273 `ORDER BY LastName ASC;`

274

275

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

StudentID	FirstName	LastName	Email
1	IsolationTest	Last1	john_updated@school.edu
10	First10	Last10	student10@school.edu
11	First11	Last11	student11@school.edu
12	First12	Last12	student12@school.edu
13	First13	Last13	student13@school.edu
14	First14	Last14	student14@school.edu
15	First15	Last15	student15@school.edu
16	First16	Last16	student16@school.edu
17	First17	Last17	student17@school.edu
18	First18	Last18	student18@school.edu

275 -- =====

276 -- STEP 17: SEARCH TEST

277 -- =====

278

279 • `SELECT * FROM Student`

280 `WHERE FirstName LIKE 'First1%';`

281

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

StudentID	FirstName	LastName	Email
10	First10	Last10	student10@school.edu
11	First11	Last11	student11@school.edu
12	First12	Last12	student12@school.edu
13	First13	Last13	student13@school.edu
14	First14	Last14	student14@school.edu
15	First15	Last15	student15@school.edu
16	First16	Last16	student16@school.edu
17	First17	Last17	student17@school.edu
18	First18	Last18	student18@school.edu

282 -- =====

283 -- STEP 18: DATE FILTER TEST

284 -- =====

285

286 • `SELECT * FROM Assignment`

287 `WHERE DueDate > '2024-03-01';`

288

289

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

AssignmentID	AssignmentName	CourseID	DueDate
1	Assignment1	5	2024-04-07
2	Assignment2	10	2024-04-09
4	Assignment4	4	2024-05-26
5	Assignment5	8	2024-03-25
6	Assignment6	8	2024-03-07
8	Assignment8	4	2024-03-16
9	Assignment9	6	2024-05-31

```
289  -- =====
290  -- STEP 19: INDEX TEST
291  -- =====
292
293  • CREATE INDEX idx_student_email ON Student(Email);
294
295  • SELECT * FROM Student
296    WHERE Email = 'student10@school.edu';
297
```

Result Grid

StudentID	FirstName	LastName	Email
10	First10	Last10	student10@school.edu